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OSSP Individual Assignment

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Section - A

Operating System - openSUSE OS

Submitted To: Lecturer Wondimu B.

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1. Introduction

In the fast-growing world of technology, open-source software plays a major role in driving innovation and supporting modern systems. One of the most important projects in this area is openSUSE, a community-based Linux distribution that is recognized for its stability, flexibility, and strong support for developers, system administrators, and everyday users.

openSUSE was first introduced by SUSE, a German company, in 1992, and over time it has developed into one of the most trusted Linux distributions worldwide. It provides two main versions to satisfy different user needs.

The first is openSUSE Leap, which is a stable long-term support (LTS) version suitable for professional and production environments. The second is openSUSE Tumbleweed, a rolling-release version that offers the latest software updates and is preferred by developers who want up-to-date tools.

The openSUSE project is managed by the openSUSE Board and supported by a global community that actively contributes to development, testing, and documentation. It focuses on openness, collaboration, and high technical standards.

Several important tools make openSUSE unique, including YaST for system configuration, Zypper for package management, Snapper for system snapshots, openQA for automated testing, and the Open Build Service for building and distributing software.

For this project, openSUSE Leap 15.5 is selected because it is a stable and well-developed platform released in 2023. It supports modern technologies such as containerization, virtualization, and advanced file systems.

Installing openSUSE in a virtual environment like VirtualBox represents real-world IT practices. Virtualization is widely used in modern infrastructure, and working in such an environment helps in developing practical skills related to system management, resource allocation, and virtual machine operation.

In conclusion, studying openSUSE is not only about learning a Linux operating system but also about understanding a powerful, enterprise-level platform that prepares learners for real-world technical challenges.

2. Objective

This project aims to build practical, technical, and professional skills through the installation, configuration, and management of openSUSE Leap 15.5 in a virtual environment. These goals are connected to essential knowledge required in IT, software engineering, system programming, and cloud computing fields.

1. **Develop Practical Skills in Linux Installation**
   You will carry out a full installation of openSUSE Leap 15.5 on a virtual machine, learning how to boot from an ISO file, set up virtual hardware components (CPU, memory, storage, and network), and resolve installation-related problems. These abilities are important for real-world system setup and deployment.
2. **Learn Linux Filesystem Structure and Partitioning**
   You will examine the Linux directory structure (such as /home, /etc, /var) and work with modern file systems like Btrfs and XFS. In addition, you will understand disk partitioning methods and how to manage system snapshots using Snapper to maintain system reliability and protect data.
3. **Build Experience with System Programming Tools**
   Within the openSUSE environment, you will work with tools like GCC, Makefiles and Bash scripting to create and execute system-level programs. This helps connect theoretical concepts with practical operating system tasks.
4. **Improve Troubleshooting and Problem-Solving Skills**
   By dealing with common issues such as virtualization errors (for example, disabled VT-x), driver problems, or network configuration errors, you will strengthen your ability to identify, analyze, and fix technical problems—skills that are highly valuable in IT careers.
5. **Understand Virtualization Technologies**
   Through installing openSUSE using tools like Oracle VM VirtualBox or VMware Workstation, you will study the basics of virtualization, including hypervisors, virtual hardware setup, resource management, and networking. This knowledge prepares you for advanced technologies such as containerization and cloud platforms.
6. **Enhance Documentation and Reporting Skills**
   You will practice writing clear and organized documentation of system installation, configuration steps, and results. This improves your ability to communicate technical processes effectively in both academic and professional environments.

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3. Requirements

Before starting the installation and setup of openSUSE Leap 15.5, it is important to clearly identify the system requirements needed for a smooth and successful process. Running an operating system in a virtual environment requires proper hardware capability and compatible software tools. Therefore, resource allocation, compatibility, and preparation must be considered carefully. This section explains the required hardware and software components.

3.1 Hardware Requirements

Processor (CPU)

A capable processor is essential for running virtual machines effectively. At minimum, a dual-core CPU with a speed of 2.0 GHz is required. For better performance, a quad-core or higher processor is recommended.

In addition, the processor must support hardware virtualization technologies such as Intel VT-x or AMD-V, and these features should be enabled in the BIOS settings. Without this support, virtualization tools like Oracle VM VirtualBox or VMware Workstation may fail to run properly and can produce errors such as “VT-x not available.”

Memory (RAM)

- Minimum: 4 GB
- Recommended: 8 GB or more

Since both the host operating system and the virtual machine run simultaneously, sufficient memory is necessary. Assigning at least 4 GB RAM to the virtual machine allows stable operation without slowdowns or crashes. For heavier tasks such as development or running multiple services, 8 GB or higher is preferred.

Storage (Disk Space)

- Minimum: 40 GB
- Recommended: 60–80 GB

Although the base installation may require around 10–15 GB, additional space is needed for software packages, updates, user files, and system snapshots. Using a

dynamically allocated virtual disk helps manage storage efficiently without consuming all physical disk space immediately.

3.2 Software Requirements

openSUSE Leap 15.5 ISO Image

The ISO file should be downloaded from the official openSUSE website to ensure it is secure and up to date. The file size is approximately 4 GB, so a stable internet connection is required.

Virtualization Software

To run the operating system, virtualization software is needed. Common options include Oracle VM VirtualBox and VMware Workstation. VirtualBox is often preferred for learning purposes because it is free, easy to use, and well-supported for Linux systems.

Terminal Emulator

After installation, a terminal interface is necessary for system management. openSUSE includes Bash by default. Users can also install Zsh for a more advanced and interactive command-line experience.

Text Editors

Text editors are required to modify configuration files and write programs. Common options include:

- Vim (advanced command-line editor)
- Nano (simple and beginner-friendly)
- Visual Studio Code (graphical editor with extensions)

Additional editors such as Emacs or Kate can also be used depending on user preference.

3.3 BIOS/UEFI Settings

Before beginning the installation process, it is necessary to enter the BIOS or UEFI settings of the host computer and configure the required options. First, hardware virtualization must be activated by enabling Intel VT-x or AMD-V, depending on the processor type.

In addition, enabling features such as IOMMU or VT-d is recommended, as they improve device management and overall system performance in a virtualized environment.

It is also advisable to set the boot mode to UEFI instead of Legacy mode, since openSUSE provides better compatibility and integration with UEFI systems.

If these configurations are not properly set, the virtual machine may fail to start or may run with very poor performance.

3.4 Network Considerations

Network configuration is another important aspect to consider when working with virtual machines. Different networking modes can be selected depending on the intended use of the virtual machine.

For general internet access, NAT mode is commonly used because it is simple and works by sharing the host system's network connection. Bridged mode, on the other hand, allows the virtual machine to act like a separate device on the same network, which is useful for real-world simulations. Internal networking can be used when communication is required only between virtual machines.

In cases where Bridged mode is used, a static IP address may be configured manually. This helps simulate real server environments and provides better control over network settings.

4. Installation Steps

Step 1: Download the openSUSE Leap 15.5 ISO

The first step is to obtain a verified, official copy of the openSUSE Leap instalation media.

1. Open a web browser and navigate to
https://download.opensuse.org/distribution/leap/15.5/iso/

2. Click on the Download Leap button. Select the x86_64 architecture, which supports 64

bit systems.

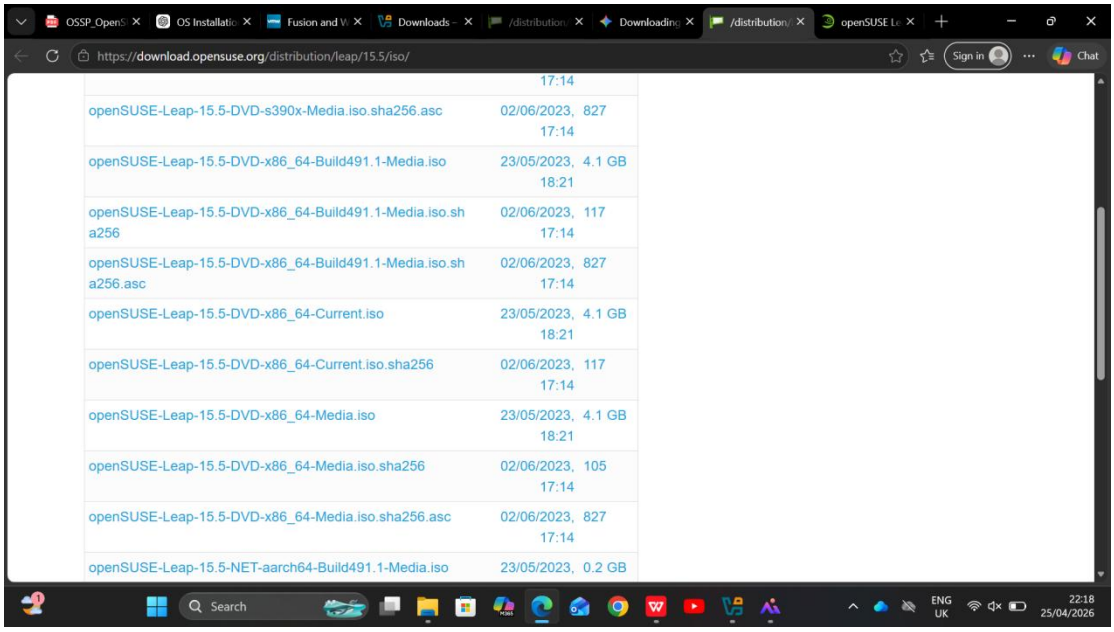
3. Choose the DVD ISO options (~4 GB size) for a complete offline installation.

4. Verify the ISO's checksum (SHA256) against the official values provided to ensure data

integrity and prevent corrupted installs.

Note: Downloading directly from openSUSE mirrors ensures you are using the latest stable

version, free from tampering or errors.



Step 2: Install and Configure VirtualBox

If not already installed, download and install the latest version of Oracle VirtualBox from

https://www.virtualbox.org/wiki/Downloads

1. Launch VirtualBox after installation.

2. Install the Extension Pack for enhaced USB support, networking options, and remote

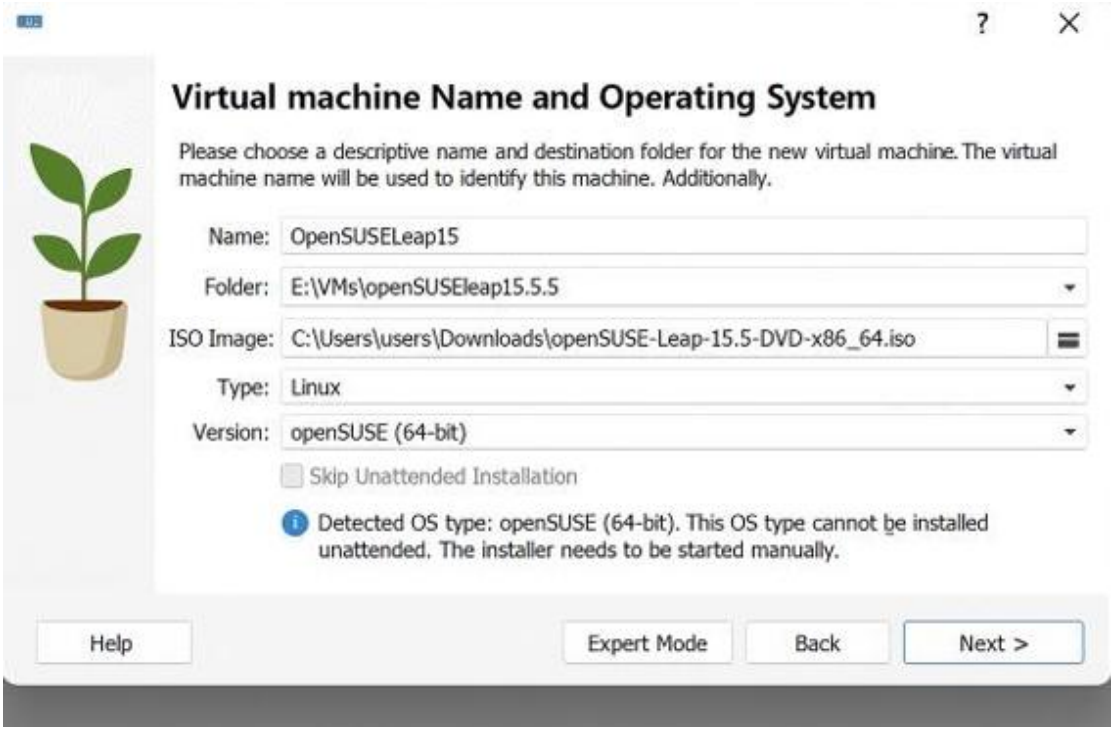
access features.

Ensure your host machine's BIOS/UEFI settings have VT-x/AMD-V virtualization support

enabled before proceding.

Step 3: Create a New Virtual Machine

1. Open Oracle VM VirtualBox and click on the **New** button to begin creating a virtual machine.
2. Provide the required details:
    1. **Name:** openSUSE (or your full name if required)
    2. **Machine Folder:** Use the default location or choose a custom directory
    3. **Type:** Linux
    4. **Version:** openSUSE (64-bit)
3. Click **Next** to continue.

4. 

Step 4: Allocate System Resources

Next, assign system resources to the virtual machine based on your computer's capacity:

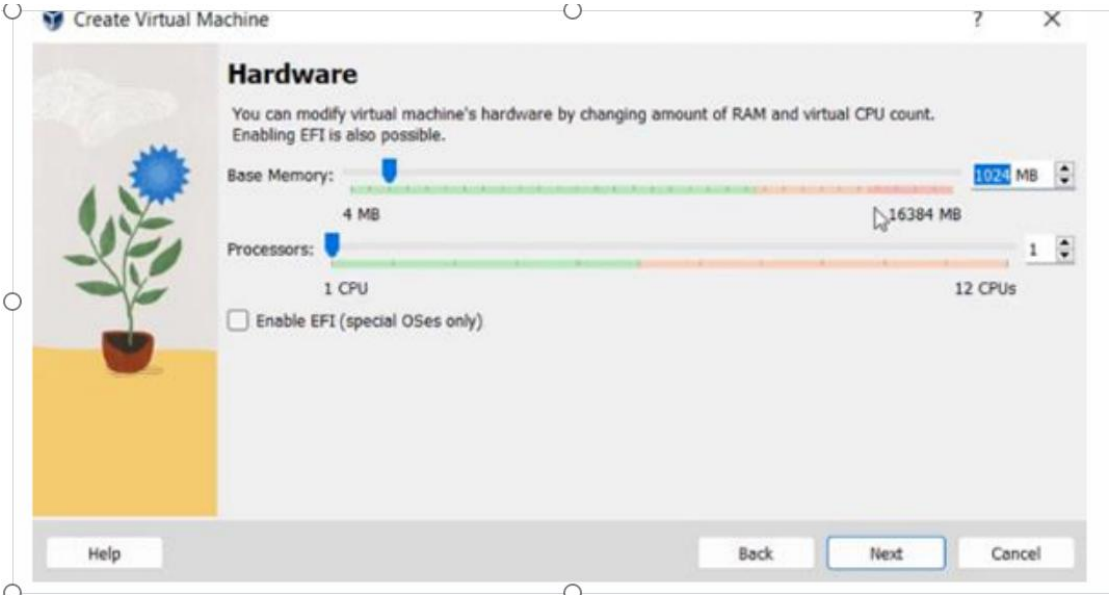
- **Memory (RAM):**
  - Minimum: 4096 MB (4 GB)
  - Recommended: 8192 MB (8 GB) for better performance
- **Hard Disk Configuration:**

  1. Select **Create a virtual hard disk now**
  2. Choose **VDI (VirtualBox Disk Image)**
  3. Select **Dynamically Allocated** storage
  4. Set disk size:
  5.

     - Minimum: 40 GB
     - Recommended: 60–80 GB if additional tools will be installed

After creating the virtual machine:

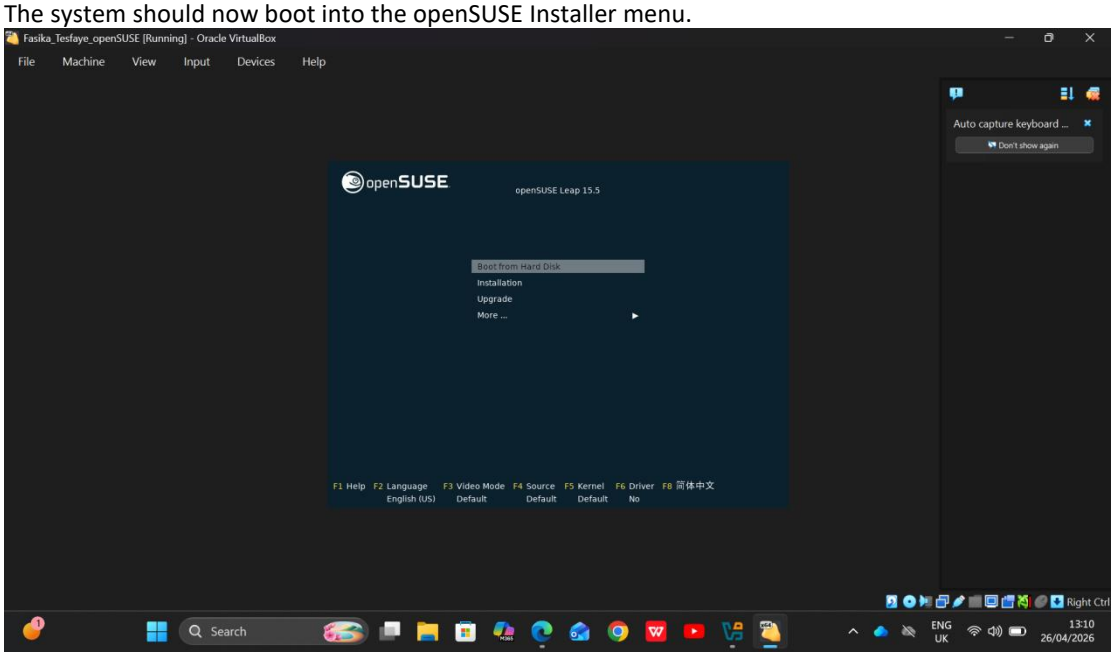
- Go to **Settings → Display**
- Increase **Video Memory** to 128 MB
- Enable **3D Acceleration** to improve graphical performance

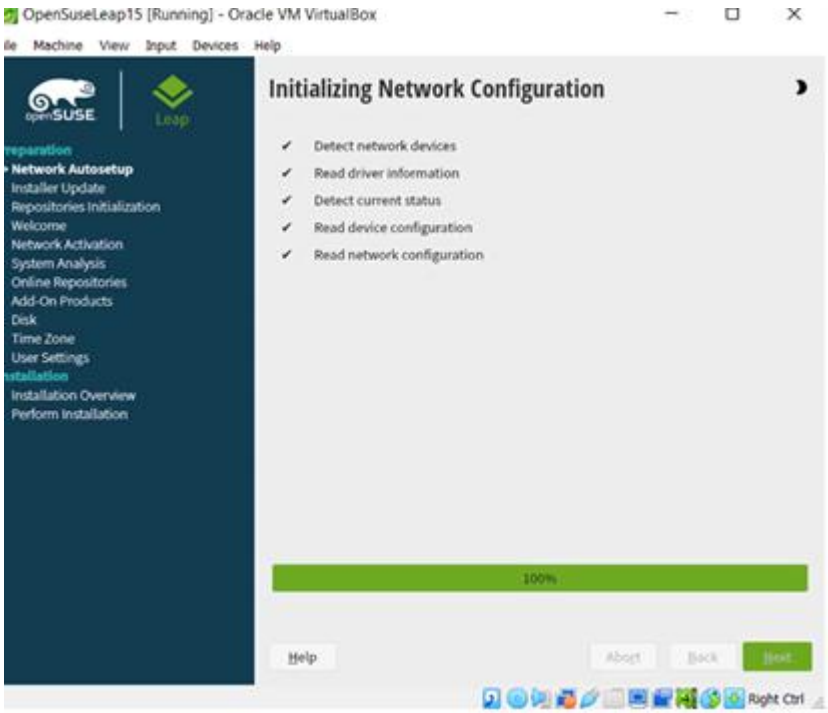


Step 5: Mount the ISO and Boot the VM

1. Start or Power on the VM.
2. Select the Installation menu.

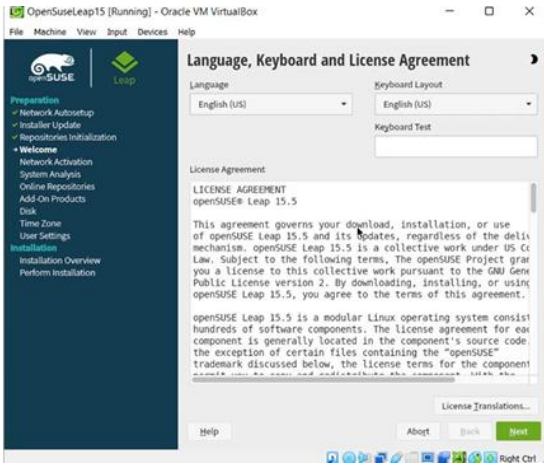
The system should now boot into the openSUSE Installer menu.

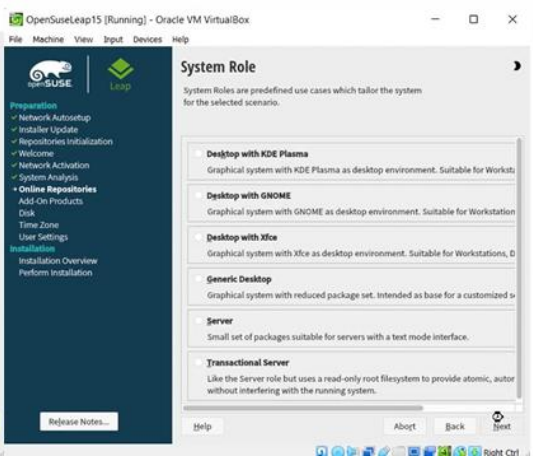




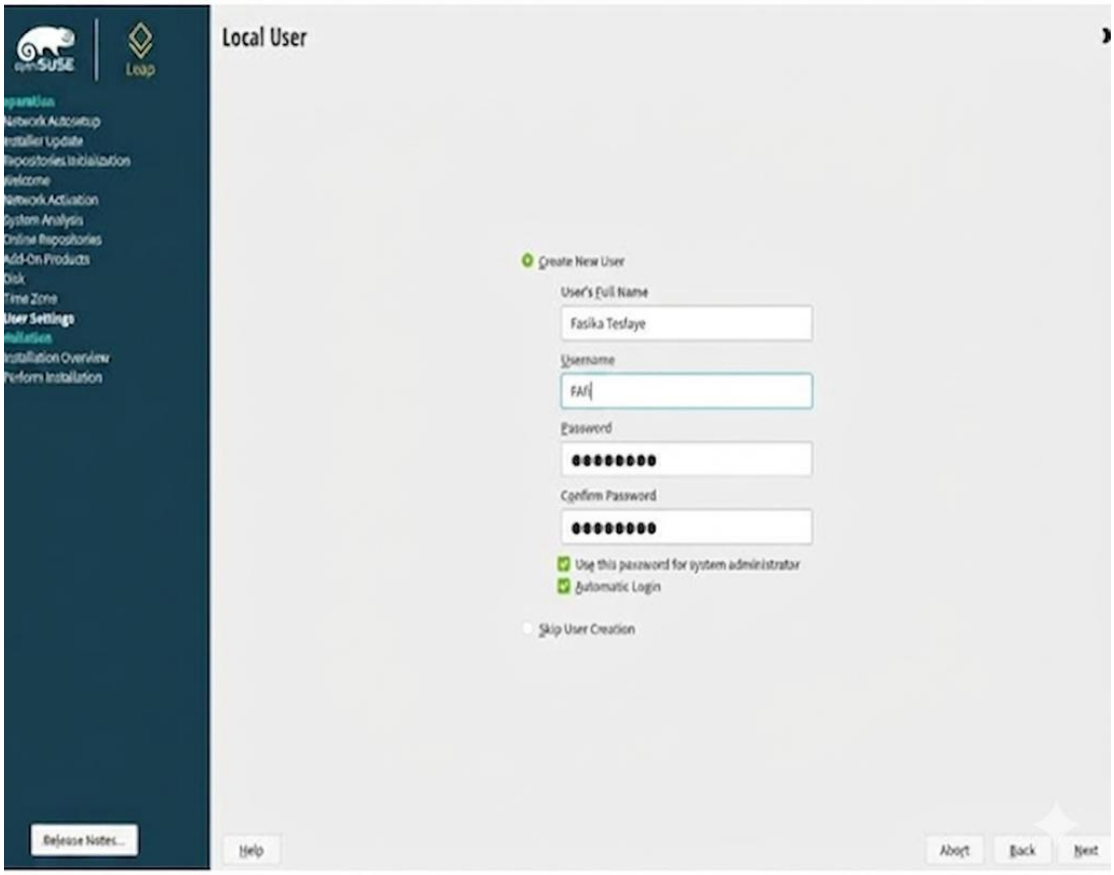
Step 6: Install openSUSE Leap 15.5

Follow these installer screens:
1. Language and Keyboard: English (US) select by default; adjust if necessary.
2. License Agreement: Read and accept the terms.
3. System Role: Choose Desktop with KDE Plasma or GNOME Desktop for a complete GUI environment.
4. Clock and Time Zone: Choose your time-zone and location.
5. Create User Account:
o Full Name: Fasika Tesfaye
o Username: FAfi
o Set a strong password.
6. Install:
o Click Install and confirm.
o The installer will proceed to copy files, install software, and configure the system.
o The process takes about 10-30 minutes depending on your hardware.


OpenSuseLeap15 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Language, Keyboard and License Agreement
Language
English (US)
Keyboard Layout
English (US)
Keyboard Test
License Agreement
LICENSE AGREEMENT
openSUSE® Leap 15.5
This agreement governs your download, installation, or use
of openSUSE Leap 15.5 and its updates, regardless of the deliv
mechanism. openSUSE Leap 15.5 is a collective work under US Co
Law. Subject to the following terms, The openSUSE Project grar
you a license to this collective work pursuant to the GNU Gene
Public License version 2. By downloading, installing, or using
openSUSE Leap 15.5, you agree to the terms of this agreement.
openSUSE Leap 15.5 is a modular Linux operating system consist
hundreds of software components. The license agreement for eac
component is generally located in the component's source code
the exception of certain files containing the "openSUSE"
trademark discussed below, the license terms for the component
License Translations...
Help
Abort
Back
Next
Right Ctrl


OpenSuseLeap15 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
System Role
System Roles are predefined use cases which tailor the system
for the selected scenario.
Desktop with KDE Plasma
Graphical system with KDE Plasma as desktop environment. Suitable for Worksta
Desktop with GNOME
Graphical system with GNOME as desktop environment. Suitable for Workstation
Desktop with Xfce
Graphical system with Xfce as desktop environment. Suitable for Workstations, D
Generic Desktop
Graphical system with reduced package set. Intended as base for a customized s
Server
Small set of packages suitable for servers with a text mode interface.
Transactional Server
Like the Server role but uses a read-only root filesystem to provide atomic, autor
without interfering with the running system.
Release Notes...
Help
Abort
Back
Next
Right Ctrl

OpenSuseLeap15 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Clock and Time Zone
Windhoek
Ethiopia
Region
Africa
Time Zone
Africa/Addis_Ababa
Other Settings...
Hardware Clock Set to UTC
2025-04-21 00:34:14
Help
Release Notes...
Abort
Next
Right Ctrl


Local User
Leap
Network Autosetup
Installer Update
Repositories Initialization
Welcome
Network Activation
System Analysis
Online Repositories
Add-On Products
Disk
Time Zone
User Settings
Installation Overview
Perform Installation
Release Notes...
Create New User
User's Full Name
Fasika Tesfaye
Username
FAFI
Password
Confirm Password
Use this password for system administrator
Automatic Login
Skip User Creation
Help
Abort
Back
Next

Step 7: Post-Installation Configuration

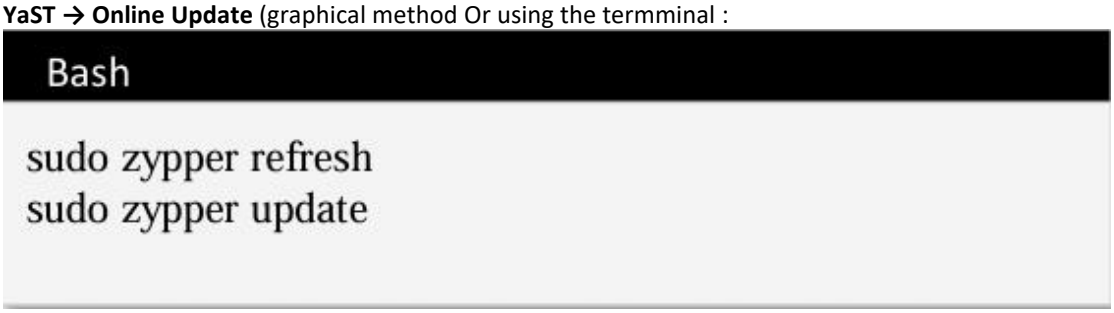
After the successful installation of openSUSE Leap 15.5, a few final steps are required to prepare the system for use.

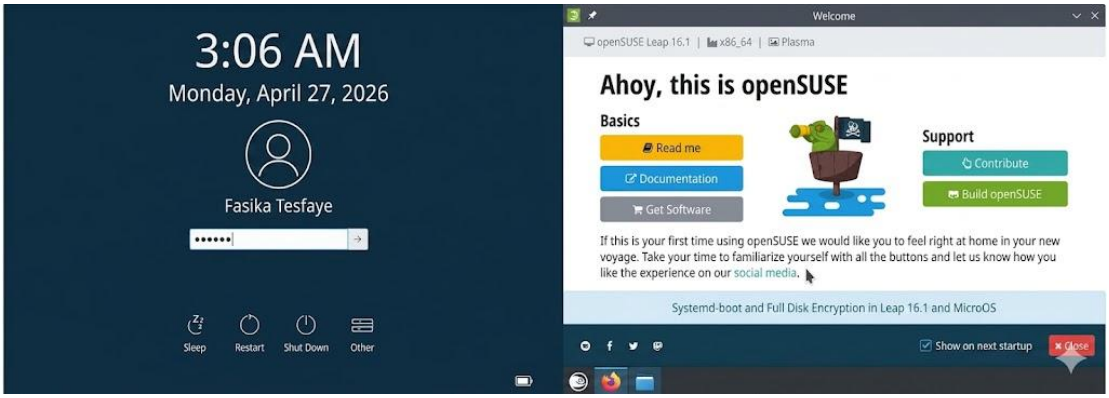
1. **System Restart**
   Once the installation process is completed, the system will restart automatically to apply all configurations.
2. **Remove Installation Media**
   After reboot, remove the ISO file from the virtual drive to prevent the system from booting into the installer again.

Go to **Device → Optical Drives → Remove Disk from Virtual Drive** in Oracle VM VirtualBox

3. **Login to the System**Enter the username and password created during installation to access the desktop environment.
4. **Update the System**It is recommended to update the system to install the latest security patches and software updates. This can be done using:

YaST → Online Update (graphical method Or using the termminal :





Step 8: Install VirtualBox Guest Additions

Guest Additions improve how your virtual machine works with your computer (host system).

1. Insert Guest Additions CD Image

- Click **Devices**
- Select **Insert Guest Additions CD Image**

This mounts a virtual CD inside your openSUSE system.

2. Open Terminal in openSUSE

```bash
sudo mount /dev/cdrom /mnt
sudo /mnt/VBoxLinuxAdditions.run
```

3. Reboot the VM

Guest Additions will enable:
Dynamic screen resizing
Shared clipboard
Better mouse pointer integration
Drag-and-drop file transfer between host and guest

5. Issues Faced

During the installation and configuration of **openSUSE Leap 15.5** in a virtualized environment, several technical challenges were encountered. These issues are listed below:

5.1 Virtualization Support Disabled (VT-x/AMD-V)

One of the most critical issues encountered at the early stage was the failure of the virtual machine to start properly, accompanied by an error message such as:

- *VT-x is not available (VERR_VMX_NO_VMX)*

This issue occurred because hardware virtualization was disabled in the system BIOS/UEFI settings. As a result, VirtualBox could not access CPU virtualization features required to run the virtual machine.

5.2 Disk Space warnings and Filesystem Alerts

During the installation of additional software packages and creation of large files, disk space warnings were displayed.

This happened due to insufficient allocated storage for the virtual hard disk, which limited the system's ability to install updates and store temporary files.

5.3 Installation Hangs During Partitioning or Package Installation

In some cases, the installation process froze or became unresponsive during partitioning or package installation. The main causes were:

- Insufficient RAM allocated to the virtual machine
- Heavy disk I/O usage shared between host and guest systems
- Background processes on the host system (such as Windows updates) consuming system resources

These factors led to reduced performance and temporary system hangs during installation.

6. Solutions

Every technical issue encountered during the installation and configuration of **openSUSE Leap 15.5** presented a valuable opportunity to develop real-world troubleshooting skills. This section explains, step-by-step, the strategies and solutions applied to resolve each problem, ensuring a successful and efficient system deployment.

6.1 Enabling Virtualization Support (VT-x/AMD-v)

To address the *VT-x not available (VERR_VMX_NO_VMX)* error, the following steps were undertaken:

1. Access BIOS/UEFI Settings:

- Restarted the host computer and entered BIOS/UEFI setup (commonly by pressing **F2, DEL, or ESC** during boot).
- Navigated to **Advanced > CPU Configuration**.

2. Enable Virtualization:

- Located **Intel Virtualization Technology (VT-x)** or **AMD-V** option.
- Changed the setting to **Enabled**.
- Saved changes and exited BIOS.

3. Disable Hyper-V (Windows Only):

- Opened **Windows Features**.
- Unchecked:
    - Hyper-V Windows Hypervisor Platform

- Restarted the system to fully disable Hyper-V, which may conflict with VirtualBox.

4. Verification:

- Relaunched VirtualBox.
- Confirmed that **64-bit guest OS options** were available.
- Successfully started the openSUSE VM without VT-x errors.

Result:

- openSUSE successfully booted and was able to fetch repositories and perform system updates.

6.2 Managing Disk Space and Snapshots

To prevent unexpected disk usage and system storage issues, the following steps were applied:

1. Monitor Snapshots:

```bash
sudo snapper list
```

Used Snapper (Btrfs snapshot tool) to view existing snapshots:

2. **Clean Unused Snapshots:**
Removed older snapshots to free space:

```bash
sudo snapper delete 1-20
```

3. **Check Disk Usage:**
Monitored storage usage using

```bash
sudo zypper clean
```

4. Expand Virtual Disk (if needed):
If storage remained low:

- Powered off the VM
- Increased virtual disk size from VirtualBox settings
- Expanded partitions inside openSUSE using system t

6.3 Handling Installation Hangs

During the installation process, the system occasionally became unresponsive or slow. To resolve this issue, the following steps were taken:

1. Allocated More Memory

- Increased the virtual machine RAM allocation from **4 GB to 6 GB**
- This improved system responsiveness during installation and package processing

2. Reduced Background Host Activity

- Closed unnecessary applications and processes on the host machine
- This freed up CPU and memory resources for the virtual machine

3. Used Text-Based Installer (If Necessary)

- From the boot menu, selected **Installation (Text Mode)**
- This reduced graphical overhead and improved installation stability in low-resource conditions

Result:

- The installation process completed successfully without further system hangs
- System stability improved significantly after resource optimization

7. Filesystem Support

A crucial aspect of any operating system's performance, stability, and resilience is its underlying filesystem. **openSUSE Leap 15.5**, being a robust and enterprise-ready Linux distribution, supports several modern filesystems, each designed to address specific use cases. This section explores these filesystems in detail, with a particular emphasis on why **Btrfs** is chosen as the default for the root partition in openSUSE installations.

7.1 Btrfs (B-tree File System)

Btrfs is the default filesystem for the root (/) partition in openSUSE Leap 15.5 installations. Originally developed by Oracle, Btrfs was designed to overcome the limitations of older filesystems like ext3 and ext4, and to meet the requirements of modern storage systems and enterprise environments.

Key Features of Btrfs:

- **Snapshot and Rollback Support:**
  Btrfs allows the creation of snapshots, which are read-only or writable copies of the filesystem at a specific point in time. In openSUSE, the **Snapper** tool is integrated with Btrfs to automatically create snapshots before system updates, enabling easy rollback in case of errors.

- **Checksumming and Data Integrity:**
  Btrfs implements end-to-end checksumming for both data and metadata. This allows detection of silent data corruption (bit rot) and improves overall system reliability.
- **Efficient Storage Management:**
  Supports advanced features such as transparent compression, deduplication, and copy-on-write (COW), which optimize disk usage and improve performance.
- **Subvolumes:**
  Instead of a traditional directory structure, Btrfs uses subvolumes. Each subvolume can be managed independently for tasks such as snapshotting, backups, and access control.
- **Online Resizing and Defragmentation:**
  Supports resizing of filesystems without unmounting and allows online defragmentation to maintain long-term performance.

why openSUSE Uses Btrfs:

- Enables seamless rollback after failed updates, improving system reliability.
- Provides strong system recovery capabilities.
- Offers advanced features without requiring complex RAID configurations.
- Aligns closely with enterprise-grade solutions through the SUSE ecosystem.

7.2 ext4 (Fourth Extended Filesystem)

ext4 remains one of the most widely used Linux filesystems and is commonly used for the **/home** partition in openSUSE installations.

Key Features of ext4:

- **High Performance:**
  Optimized for fast operations and minimal overhead in everyday use.
- **Large Volume Support:**
  Supports filesystems up to 1 exabyte (EB) and individual files up to 16 terabytes (TB).
- **Backward Compatibility:**
  Fully compatible with older ext3 and ext2 filesystems.
- **Stability and Maturity:**
  Extensively tested and widely deployed across millions of Linux systems worldwide.

In openSUSE, ext4 is often selected manually by users who prefer a traditional filesystem without snapshot or advanced management features.

7.3 XFS

XFS is a high-performance journaling filesystem originally developed by SGI (Silicon Graphics Inc.) for high-throughput computing environments.

Key Features of XFS:

- **Excellent Parallelism:**
  Efficiently handles simultaneous read and write operations, making it ideal for server workloads.
- **High Scalability:**
  Designed to support extremely large files and a very large number of files without performance degradation.
- **Dynamic Allocation:**
  Uses intelligent space allocation techniques to reduce fragmentation and optimize storage efficiency.

openSUSE includes XFS as an option during manual partitioning, particularly recommended for directories such as **/var** or **/srv**, where heavy file input/output (I/O) activity is expected.

7.4 FAT32 and NTFS

Although not native Linux filesystems, **FAT32** and **NTFS** are supported in openSUSE through appropriate drivers (such as *ntfs-3g*) to ensure interoperability with Windows systems.

FAT32:

- Commonly used for USB flash drives and external storage devices
- Compatible with almost all operating systems
- Limited by a maximum file size of **4 GB**, which restricts usage for large files

NTFS:

- Default filesystem for Windows operating systems
- Supports large files, along with features such as encryption, compression, and journaling
- Provides full read/write access in Linux through the **ntfs-3g** driver

In openSUSE, external storage devices formatted with FAT32 or NTFS are automatically detected and mounted using graphical file managers such as **Dolphin (KDE)** or **Nautilus (GNOME)**, ensuring seamless user interaction.

7.5 Default Partitioning Scheme in openSUSE Leap 15.5

When the guided partitioning option is selected during installation, openSUSE typically proposes the following default layout:

- **/** (root) → formatted with **Btrfs**
- **/home** → formatted with **XFS** or **ext4**
- **Swap area** → allocated based on system memory (typically **2 GB – 8 GB**)

This default configuration is designed to balance **system stability and performance**:

- Btrfs enables snapshot and rollback functionality for the root system
- XFS or ext4 provides efficient performance for user data in the /home directory

- Swap space supports memory management and system responsiveness

8. Advantages and Disadvantages

Every operating system, no matter how mature and well-designed, has both strengths and weaknesses depending on its target user base, use cases, and technical objectives. **openSUSE Leap 15.5**, developed and maintained by the openSUSE community in collaboration with SUSE, is no exception. This section analyzes the key advantages and disadvantages of openSUSE Leap based on firsthand experience, official documentation, and community feedback.

8.1 Advantages

1. Enterprise-Grade Stability

openSUSE Leap shares its codebase with **SUSE Linux Enterprise (SLE)**, a commercial, enterprise-level Linux distribution. As a result, Leap benefits from rigorous quality control, strong security practices, and long-term support.

Users gain a high level of system stability and reliability, which is essential for:

- Professional servers
- Development environments
- Personal computing

Example: Leap users receive security patches and bug fixes that have been thoroughly tested in enterprise environments, reducing downtime and operational risks.

2. Advanced System Management via YaST

YaST (Yet Another Setup Tool) is one of openSUSE's most powerful and distinctive features. It provides both a graphical user interface (GUI) and a command-line interface (CLI) for system administration.

Through YaST, users can:

- Partition disks
- Configure network interfaces
- Manage users and services
- Install or remove software packages
- Configure firewalls, NTP, and security policies

YaST simplifies complex administrative tasks that, in many other Linux distributions, require manual editing of configuration files.

3. Snapshot and Rollback Capabilities (Btrfs + Snapper)

By using **Btrfs** as the default root filesystem, openSUSE Leap integrates seamlessly with **Snapper**, a snapshot management tool.

Key benefits include:

- **Automatic Snapshots:** Created before major system updates
- **System Rollback:** Users can restore a previous system state via the GRUB boot menu
- **Space Efficiency:** Copy-on-write (COW) reduces disk usage overhead

This feature provides reliability and confidence when performing system updates or configuration changes.

4. Strong Security and Access Control

openSUSE emphasizes security through multiple built-in mechanisms:

- **AppArmor:** A Mandatory Access Control (MAC) system enabled by default
- **Open Build Service (OBS):** Ensures secure and verified package builds
- **Timely Security Updates:** Vulnerabilities are addressed quickly with well-tested patches

Additionally, openSUSE follows system hardening practices aligned with industry standards such as CIS Benchmarks.

5. wide Software Availability

openSUSE offers access to a large and diverse software ecosystem through:

- Official repositories
- Community repositories (e.g., Packman)
- Open Build Service (OBS)

Available software includes:

- Programming languages (Python, C++, Rust, Go)
- Web servers (Apache, Nginx)
- Databases (MySQL/MariaDB, PostgreSQL)
- Desktop environments (KDE Plasma, GNOME, XFCE, LXQt)

If specific software is not readily available, users can build and customize packages using OBS.

6. Transparent Community Governance

The openSUSE project is governed transparently by the **openSUSE Project Board**, whose members are elected by the community. Major decisions are made through open discussions and community voting, ensuring a democratic and inclusive development process.

This governance model promotes:

- Trust and transparency
- Community participation
- Continuous innovation

Although the community-driven model may sometimes result in slower decision-making, the availability of strong documentation and active community forums helps to compensate for this limitation.

8.2 Disadvantages

1. Higher System Resource Requirements

Compared to lightweight Linux distributions such as Lubuntu or minimal installations of Arch Linux, openSUSE Leap requires relatively more system resources, including RAM, CPU, and storage.

This is due to:

- Enterprise-level features
- Automatic snapshot functionality (Btrfs + Snapper)
- Comprehensive graphical tools like YaST

Example: A base KDE installation may consume approximately **700–900 MB of RAM at idle**, which is higher than many lightweight distributions.

This can be a disadvantage for users running openSUSE on older or low-spec hardware.

2. Learning Curve for Beginners

While YaST simplifies many administrative tasks, openSUSE still includes advanced features that require a deeper level of understanding, such as:

- Btrfs filesystem management
- Snapper configuration
- systemd service management
- OBS (Open Build Service) package creation

As a result, beginners may find openSUSE more complex compared to more beginner-friendly distributions like Ubuntu or Linux Mint.

3. Smaller Desktop market Share

Although openSUSE is widely respected in enterprise and professional environments, its desktop user base is smaller compared to distributions such as Ubuntu or Fedora.

This can lead to:

- Fewer tutorials and learning resources specifically tailored to openSUSE
- Slower adoption of certain desktop applications optimized for other distributions

However, the strong openSUSE community and official documentation help mitigate these challenges.

4. Complexity of Snapper and Btrfs Management

While **Snapper** and **Btrfs** provide powerful snapshot and rollback capabilities, they also introduce a level of complexity:

- Snapshot storage must be monitored and managed to prevent excessive disk usage
- Btrfs subvolume structures require deeper understanding for advanced configuration
- Improper setup may lead to confusing disk space warnings or reduced system performance

For inexperienced users, these advanced features can become difficult to manage without proper guidance.

8.3 Summary of Advantages and Disadvantages

Advantages	Disadvantages
Enterprise-grade stability	Higher system resource usage
Powerful YaST administration	Steep learning curve
Reliable snapshot and rollback	Smaller desktop community
Strong security framework	Complexity of Btrfs and snapshots
Large software ecosystem	

9. Conclusion

The installation, configuration, and exploration of **openSUSE Leap 15.5** within a virtualized environment, as part of this OSSP (Operating Systems and System Programming) project, provided a comprehensive and valuable learning experience. It

effectively combined theoretical knowledge with hands-on technical practice, aligning well with the academic objective of developing practical skills in system programming, administration, and modern IT infrastructure management.

Through this project, real-world scenarios commonly encountered in professional environments were explored. These included managing hardware virtualization, performing Linux installations, working with different file system architectures, troubleshooting system-level issues, and optimizing system performance. This experience goes beyond basic Linux usage and contributes to building a strong systems engineering mindset.

Mastery of Practical Skills

The step-by-step process of downloading the openSUSE ISO, configuring the virtual machine, installing the operating system, and setting up essential services closely simulated real-world system administration tasks. Activities such as memory allocation, disk configuration, and network setup reinforced best practices in system deployment and management.

Additionally, resolving challenges—such as enabling virtualization (VT-x/AMD-V), installing Virtual Box Guest Additions, and fixing network-related issues—helped develop critical thinking and problem-solving skills essential for IT professionals.

Deepening Understanding of openSUSE

Working with openSUSE Leap 15.5 provided insight into an enterprise-oriented Linux distribution focused on stability, security, and advanced system management. Tools such as **YaST**, **Snapper**, and the **Open Build Service (OBS)** demonstrate openSUSE's commitment to flexibility, reliability, and transparency.

Exploring file systems such as **Btrfs**, **XFS**, and **ext4**, along with practical experience in snapshot and rollback management, enhanced understanding of system recovery and data protection—key aspects in modern computing environments.

Understanding Modern Computing Practices

The use of virtualization platforms such as Virtual Box reflects modern industry practices, where physical hardware is abstracted into virtual environments. This knowledge is essential for careers in cloud computing, containerization, orchestration (e.g., Kubernetes), and distributed systems.

Furthermore, exposure to UNIX standards (such as POSIX) and basic system programming concepts (including system calls and shell scripting) helped bridge the gap between user-level operations and core operating system functionality.

Professional Development

This project highlighted the importance of structured problem-solving, clear documentation, independent research, and continuous learning. It encouraged a deeper understanding of system behavior rather than simply following instructions, promoting a more professional and analytical approach to computing tasks.

Future Relevance

The skills developed through this project—including Linux system administration, filesystem management, virtualization, scripting, and troubleshooting—are highly relevant across multiple career paths such as software development, cybersecurity, cloud engineering, and IT support.

With Linux continuing to dominate server and cloud environments, and virtualization forming the backbone of modern infrastructure, the knowledge gained from this project provides a strong foundation for future professional growth.

Future Outlook / Recommendations

While the successful installation and configuration of **openSUSE Leap 15.5** marks a significant achievement, it also serves as a foundation for deeper technical exploration and professional development.

Beyond basic system setup, users are encouraged to explore **openSUSE MicroOS**, a modern variant designed for cloud-native and edge computing environments. It introduces transactional and atomic updates along with reliable rollback mechanisms—features that are essential in modern DevOps workflows. Understanding this model provides valuable insight into maintaining system stability in dynamic deployment scenarios.

To further enhance system administration skills, students should explore advanced **Bash scripting** and automation techniques. Writing scripts that include error handling, logging, and task scheduling (via cron jobs) promotes efficiency and real-world problem-solving. Additionally, learning configuration management tools such as **Ansible**, **Puppet**, or **SaltStack** enables the implementation of **Infrastructure as Code (IaC)**, aligning with modern enterprise and cloud practices.

The **YaST** toolset also offers opportunities for deeper learning in areas such as:

- Virtual machine management

- Snapshot configuration
- User and security administration
- Bootloader and system services management

This expands both administrative control and system-level understanding.

Furthermore, deploying real-world services on openSUSE—such as:

- Web servers (**Apache**, **Nginx**)
- Remote access (**OpenSSH**)
- File sharing (**Samba**)
- Network services (**DNS**, **DHCP**)

provides hands-on experience in networking and server administration.

Active participation in the open-source community—through documentation, package contributions, or bug reporting—can further enhance both technical and collaborative skills.

This foundation also prepares learners for industry certifications such as:

- LPIC (Linux Professional Institute Certification)
- CompTIA Linux+
- SUSE Certified Administrator (SCA)
- RHCSA (Red Hat Certified System Administrator)

Finally, further study in:

- Containerization (**Docker**, **Podman**)
- Virtualization (**KVM**, **QEMU**)
- Orchestration (**Kubernetes**)
- Security frameworks (**SELinux**, **AppArmor**)

will strengthen expertise in enterprise and cloud-based technologies.

10. Virtualization in Modern Operating Systems

Virtualization is one of the most significant technological advancements in modern computing. It fundamentally transforms how software interacts with hardware, how IT infrastructures are designed, and how services are deployed at scale. Understanding virtualization is essential for software engineers, system administrators, cloud architects, and DevOps professionals.

This section provides an overview of virtualization concepts, types, technologies, and their integration within modern operating systems, particularly in Linux environments such as openSUSE.

10.1 What is Virtualization?

Virtualization refers to the creation of a virtual (rather than physical) version of computing resources, including hardware platforms, storage devices, networks, and operating systems.

Instead of relying solely on physical hardware, virtualization uses software to simulate hardware functionality. This enables multiple operating systems to run simultaneously on a single physical machine, with each instance operating independently in an isolated environment.

10.2 Types of Virtualization Virtualization can be classified based on the resources being virtualized:

a) Hardware Virtualization

- Simulates complete physical machines
- Each Virtual Machine (VM) operates as an independent system
- Examples: VirtualBox, VMware, KVM

b) Operating System-Level Virtualization (Containers)

- Virtualizes at the OS level
- Containers share the host OS kernel but maintain isolated user environments
- Technologies: Docker, Podman, LXC

c) Storage Virtualization

- Combines multiple physical storage devices into a single logical unit
- Improves performance, scalability, and redundancy
- Technologies: RAID, LVM, Ceph

d) Network Virtualization

- Abstracts physical network resources into software-defined components
- Examples include VLANs and Software-Defined Networking (SDN)

10.3 Hypervisors

Hypervisors, also known as **Virtual Machine Monitors (VMMs)**, are responsible for creating and managing virtual machines.

Type 1 (Bare-metal Hypervisors)

- Run directly on physical hardware
- Examples: VMware ESXi, Microsoft Hyper-V, KVM
- Commonly used in data centers and cloud environments

Type 2 (Hosted Hypervisors)

- Run on top of an existing operating system
- Examples: Oracle VirtualBox, VMware Workstation
- Ideal for desktop use, testing, and learning environments

openSUSE Leap supports **KVM-based virtualization**, allowing systems to function as powerful, server-grade hypervisors.

10.4 Advantages of Virtualization

- **Resource Optimization:** Reduces hardware costs by running multiple systems on a single machine
- **Isolation:** Failures in one VM do not affect others
- **Scalability:** Quickly deploy new virtual environments
- **Disaster Recovery:** Simplifies backup and restoration processes
- **Testing and Development:** Provides safe environments for experimentation without affecting production systems

10.5 Virtualization in Modern openSUSE Systems

openSUSE Leap 15.5 provides extensive support for a wide range of virtualization technologies:

- **KVM and QEMU:** Native support for hardware-accelerated virtual machines
- **libvirt and virt-manager:** Tools for managing virtual machines through both CLI and GUI
- **Xen Hypervisor:** Supported for high-performance virtualization environments
- **Docker, Podman, Kubernetes:** Comprehensive support for containerization and orchestration
- **YaST Virtualization Module:** Simplifies the configuration and management of hypervisors and guest systems

The integration of these technologies enables users to build hybrid environments that combine virtual machines and containers, reflecting modern enterprise infrastructure design

.

10.6 Virtualization vs. Containerization

Aspect	Virtualization	Containerization
Overhead	High (full OS per VM)	Low (shared host OS kernel)
Isolation Level	Full hardware-level isolation	Process-level isolation
Performance	Slightly lower due to hypervisor overhead	Near-native performance
Startup Time	Slower (requires OS boot)	Fast (containers start in seconds)
Use Cases	Multi-OS environments, legacy applications	Microservices, cloud-native applications

Understanding the distinction between virtual machines and containers is essential for designing efficient and scalable systems in modern cloud and enterprise environments.

10.7 Future of Virtualization

The future of virtualization is evolving toward more lightweight, scalable, and automated models:

- **Serverless Computing:** Eliminates the need to manage servers directly
- **Edge Computing:** Deploys workloads closer to end users for reduced latency
- **Hyperconverged Infrastructure (HCI):** Integrates compute, storage, and networking into unified systems
- **MicroVMs:** Lightweight virtual machines (e.g., Firecracker) designed for fast startup and minimal overhead

Linux distributions such as openSUSE continue to evolve by enhancing support for container orchestration platforms like Kubernetes, hybrid cloud environments, and optimized virtualization stacks.

Conclusion on Virtualization

Virtualization has become a foundational technology in modern IT environments. It enables organizations to efficiently utilize resources, isolate workloads, and deploy scalable systems.

Whether used for running multiple operating systems, deploying containerized applications, or creating secure testing environments, virtualization enhances flexibility, resilience, and cost-effectiveness.

For students and IT professionals alike, developing a strong understanding of virtualization concepts—along with hands-on experience, as demonstrated in this OSSP project—is essential for building a successful career in today's technology-driven world.

11. System Call Implementation Example: `fdatasync()`

11.1 User Space (glibc wrapper)

When you call `fdatasync(fd)` in a C program, you're not directly calling the kernel. Instead, you're using a wrapper provided by the GNU C Library (glibc):

```c
#include <unistd.h>
#include <sys/syscall.h>
#include <errno.h>

int fdatasync(int fd) {
    return syscall(SYS_fdatasync, fd);
}
    return ret;
}
```

This wrapper just use the syscall( )functione to invoke the actual kernel system call

11.2 Kernel Space: The Actual System Call

In the Linux kernel, the implementation of `fdatasync()` is located in the `fs/sync.c` file. It is defined using the system call macro as follows:

```c
SYSCALL_DEFINE1(fdatasync, int, fd)
{
    struct fd f = fdget(fd);
    int ret = -EBADF;

    if (f.file) {
        ret = vfs_fsync(f.file, 1);
        fdput(f);
    }

    return ret;
}
```

Here's what happens:

- fdget(fd) – Gets the struct file * for the file descriptor.
- vfs_fsync(file, datasync) – Calls the VFS (Virtual File System) sync function.
- The second argument 1 indicates it's a **data-only** sync (fdatasync), not a full sync.
- fdput(f) – Releases the file descriptor reference.

11.3 Under the hood: vfs_fsync

```c
int vfs_fsync(struct file *file, int datasync)
{
    if (!file->f_op->fsync)
        return -EINVAL;

    return file->f_op->fsync(file, file-
>f_path.dentry, datasync);
}
```

This delegates to the file system's ***fsync*** method (***ext4***, ***xfs***, etc.), passing along the ***datasync*** flag. File systems will then decide how to flush data and whether to skip metadata.

Summary

Layer	Function	Role
User space	fdatasync(int fd)	glibc wrapper that calls syscall()
Syscall entry	SYSCALL_DEFINE1(fdatasync)	Kernel syscall interface
VFS layer	vfs_fsync()	Generic sync logic
File system	ext4_fsync(), etc.	Filesystem-specific implementation

References

openSUSE Official Sources

- openSUSE Main Portal: https://www.opensuse.org/
- openSUSE Leap Portal: https://en.opensuse.org/Portal:Leap
- openSUSE Documentation: https://doc.opensuse.org/
- openSUSE Wiki: https://en.opensuse.org/Main_Page

Virtualization Resources

- Oracle VirtualBox Official Website: https://www.virtualbox.org/
- **Additional Learning Resources**
- openSUSE Build Service (OBS): https://build.opensuse.org/
- openQA Testing Framework: https://open.qa/